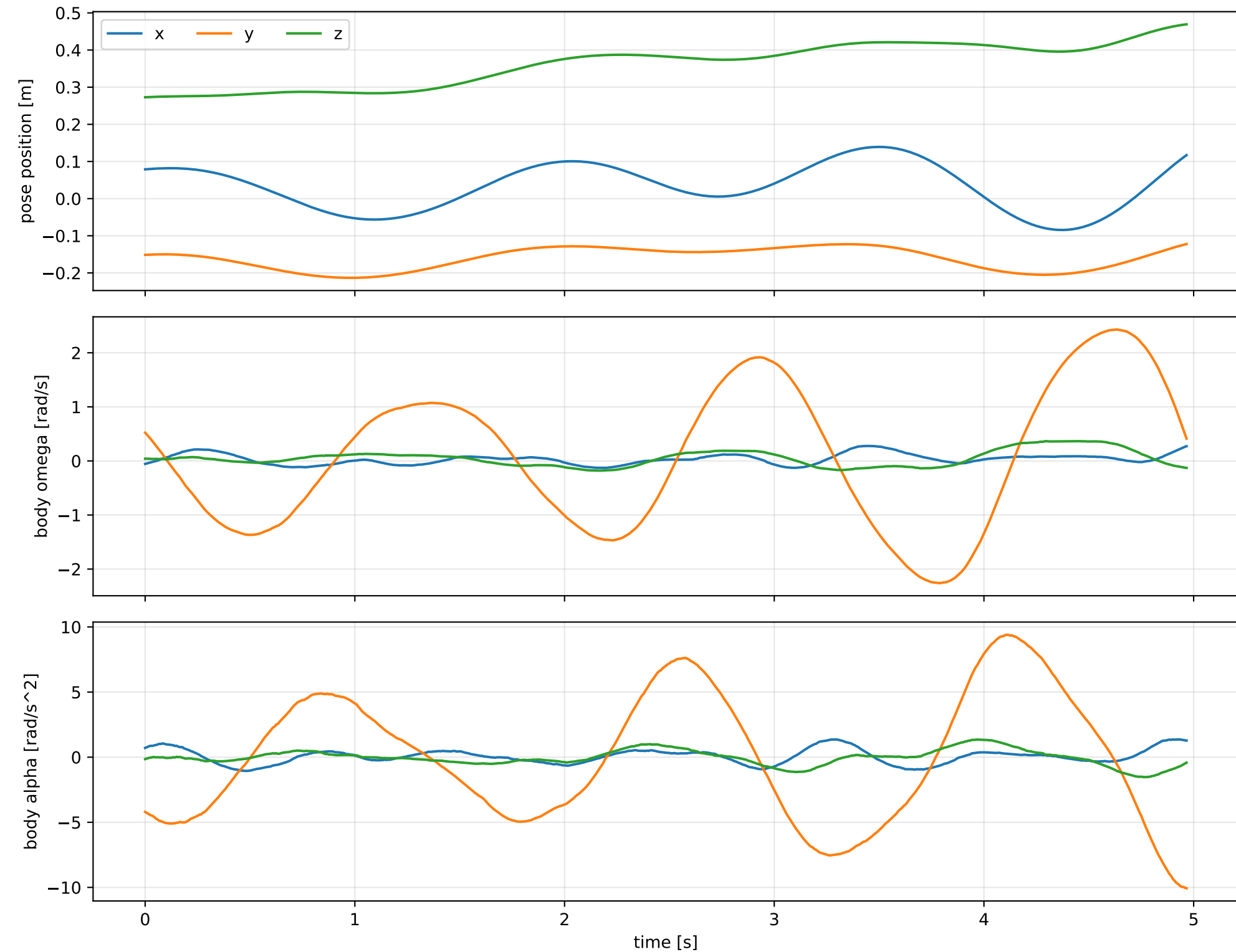
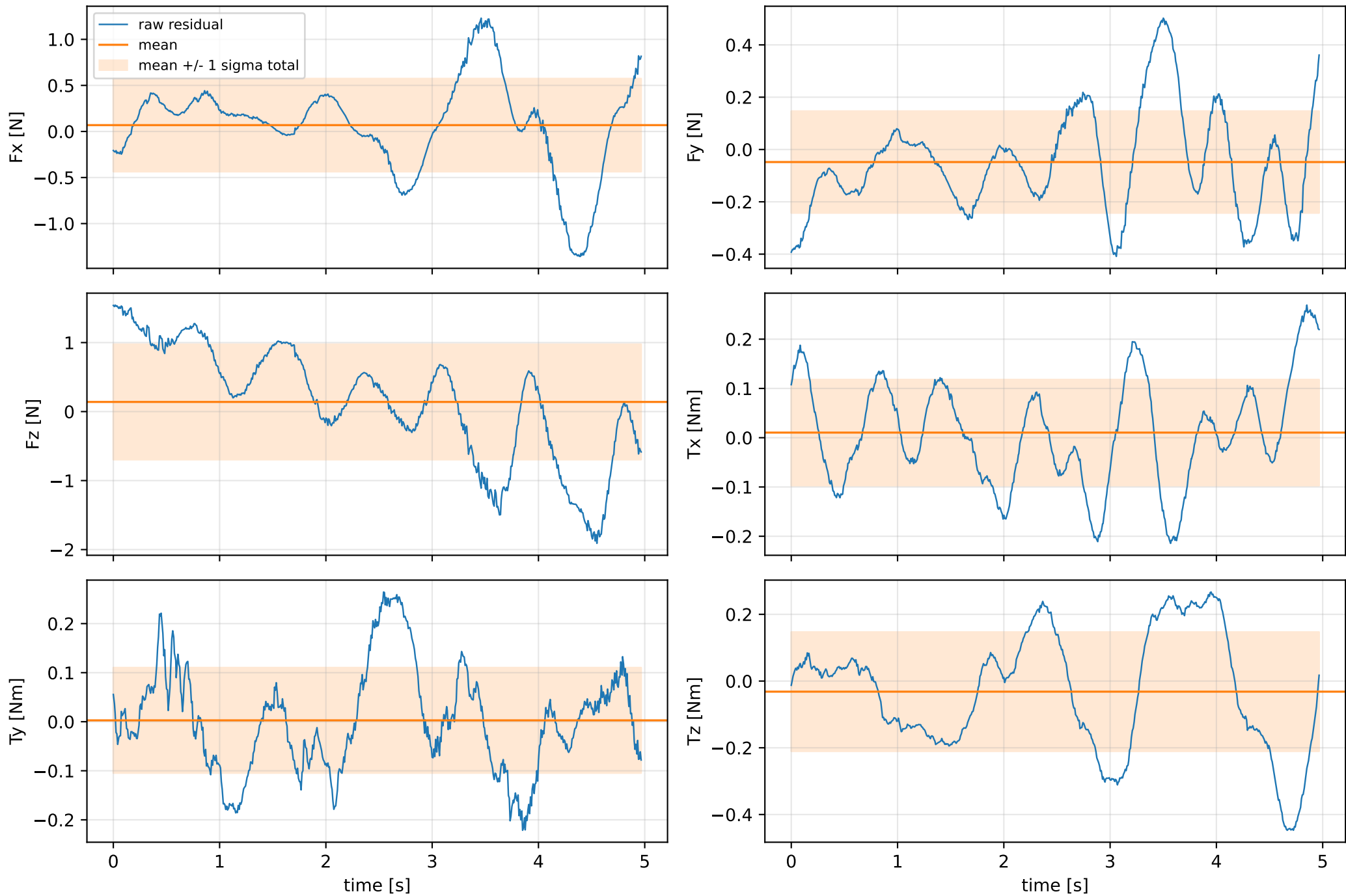


cog\_x\_nominal: SG trajectory and derived motion



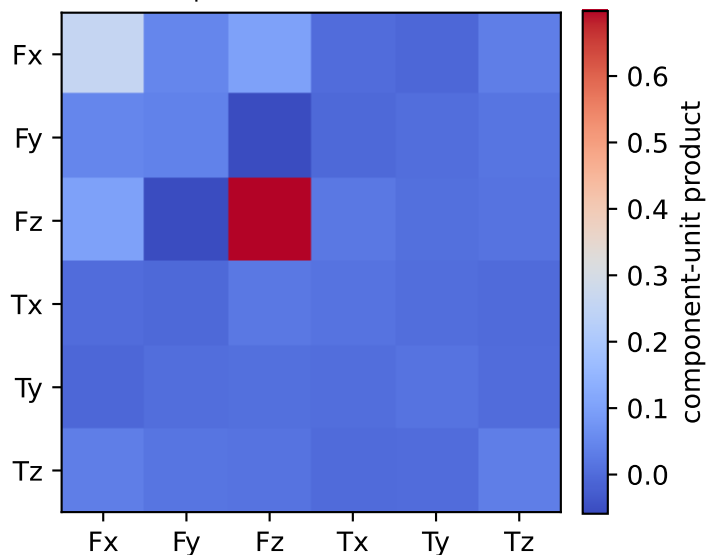
cog\_x\_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)  
mass gauge fixed to nominal mass = 2.35159759 kg



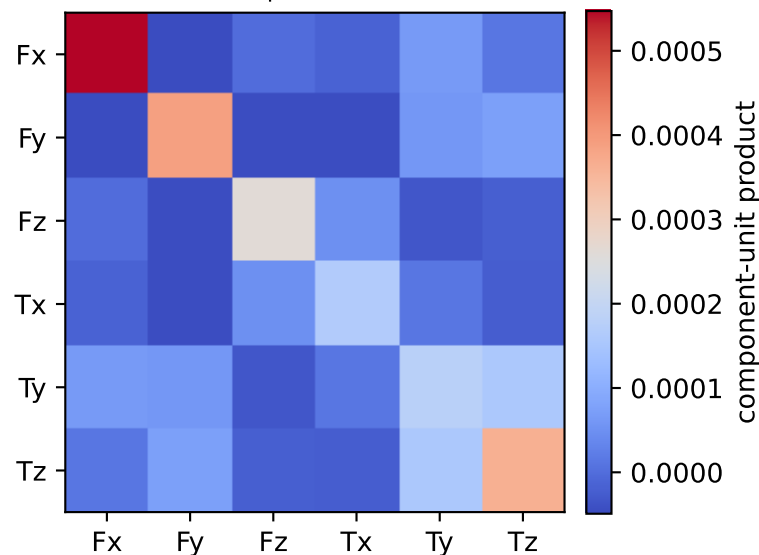
vector RMS about zero: force 1.0076 N, torque 0.23666 Nm; component std about mean: force [0.5046 0.1945 0.8354], torque [0.1075 0.1074 0.1786]

cog\_x\_nominal: residual-wrench covariance decomposition in nominal-mass gauge  
force [N], torque [Nm]; raw excess eigenvalues [0.008 0.009 0.018 0.032 0.253 0.723]

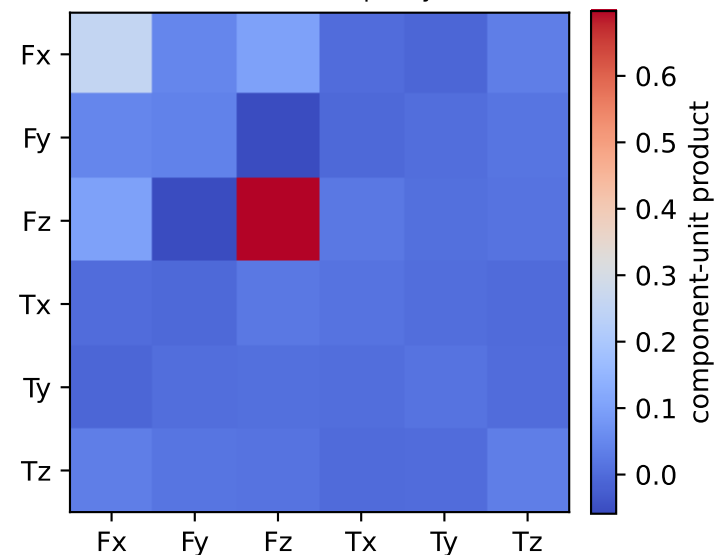
empirical total covariance



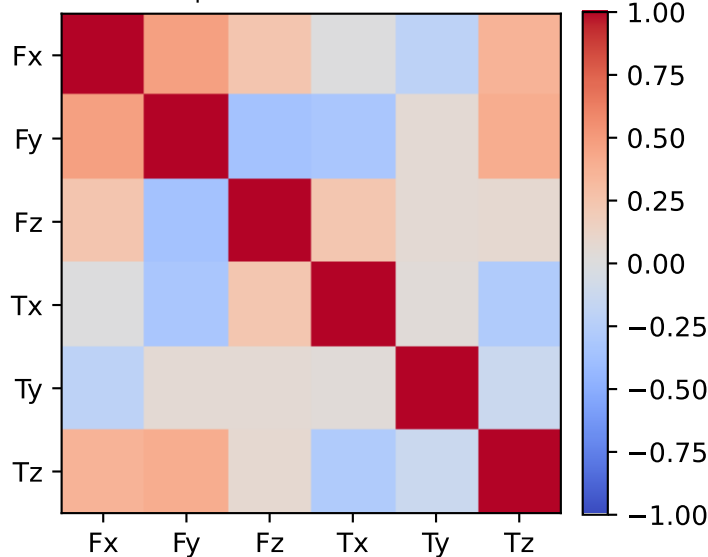
mean full-SG prediction covariance



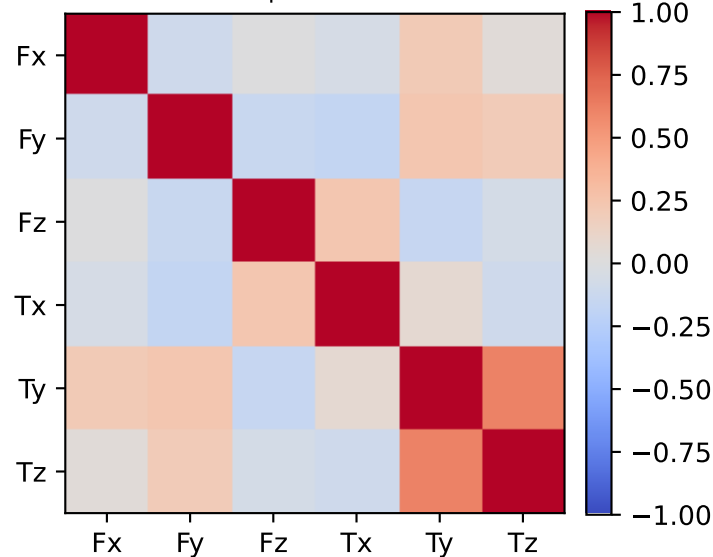
PSD excess model discrepancy covariance



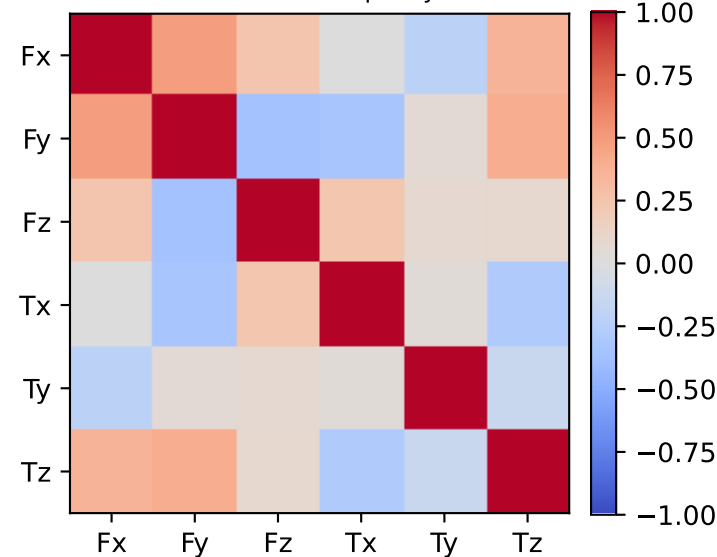
empirical total correlation



mean full-SG prediction correlation

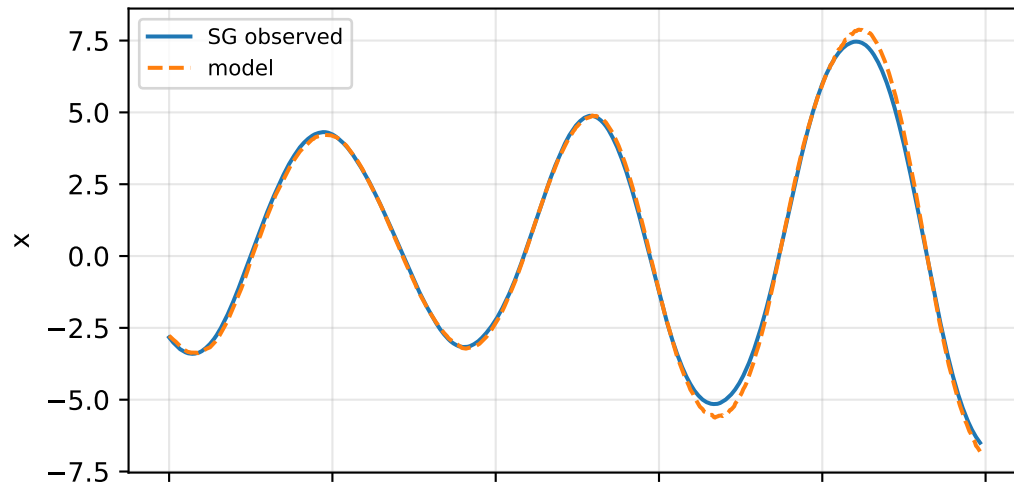


PSD excess model discrepancy correlation

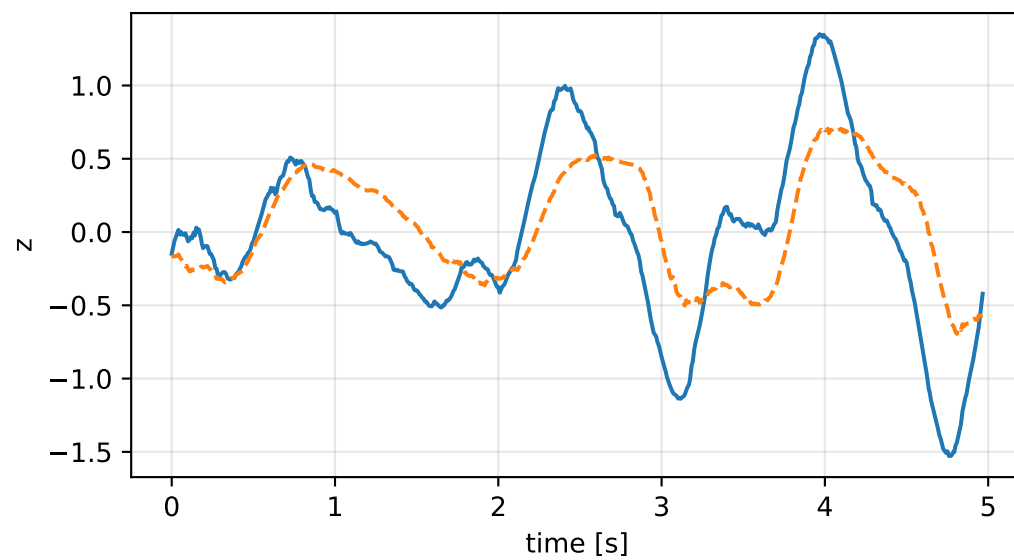
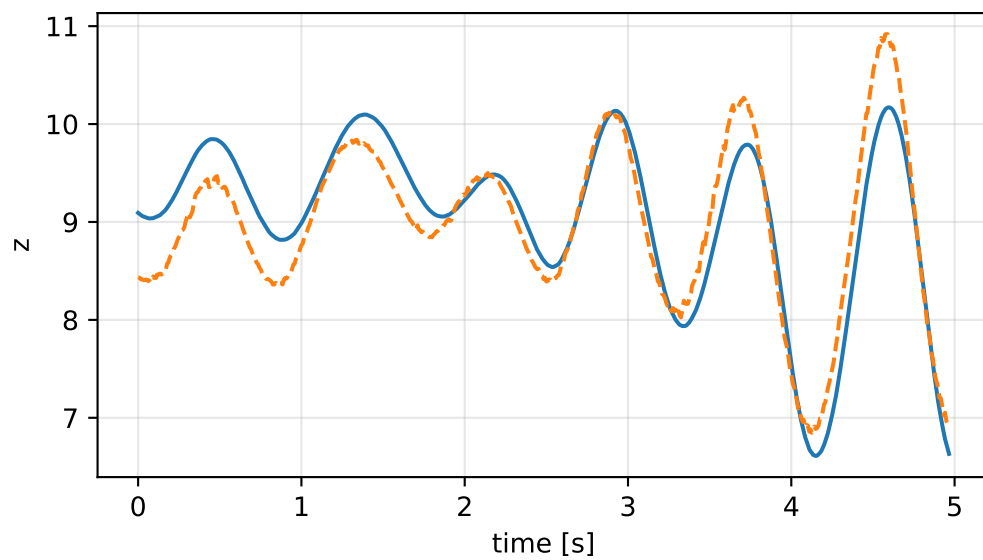
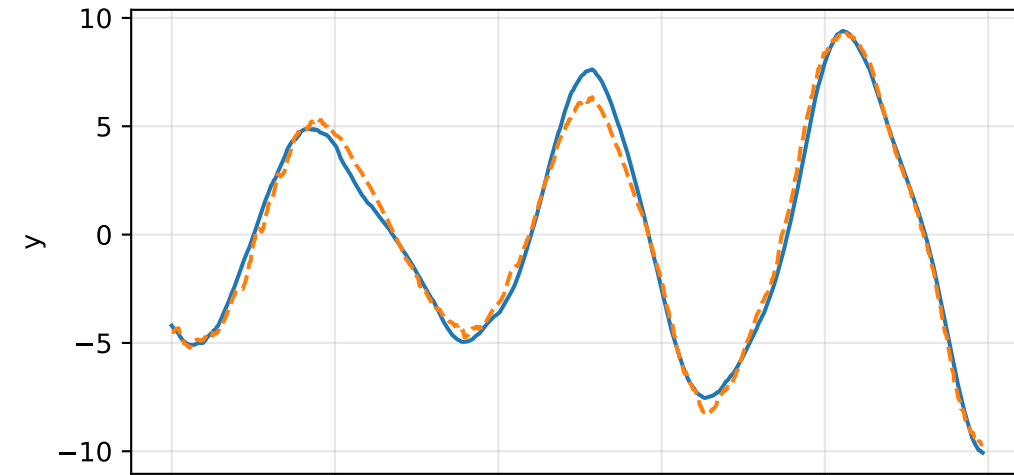
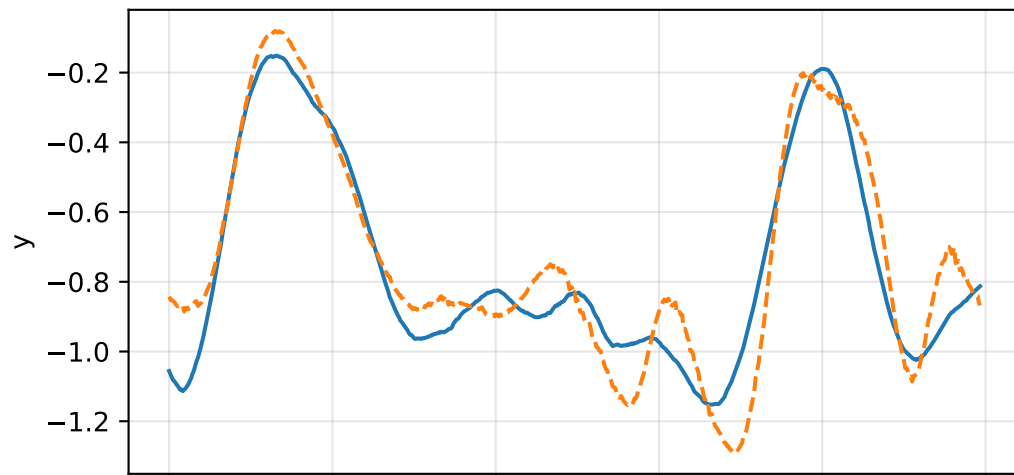
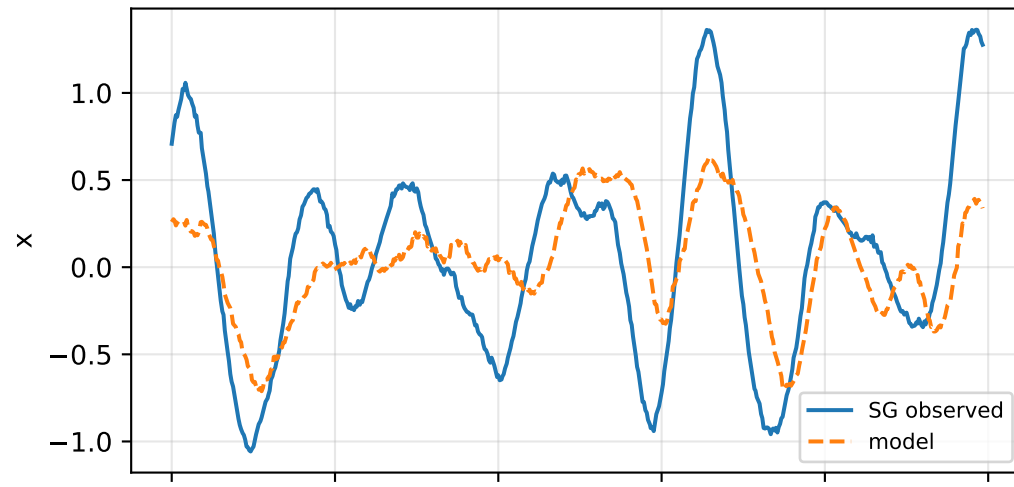


# cog\_x\_nominal: acceleration residual objective

## specific acceleration [m/s<sup>2</sup>]

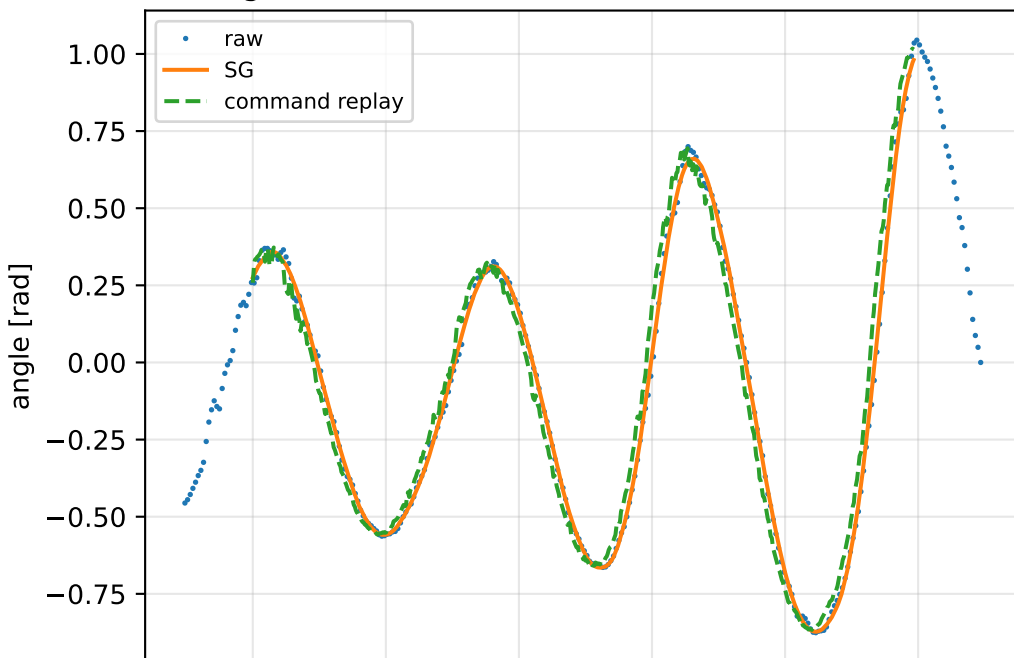


## angular acceleration [rad/s<sup>2</sup>]

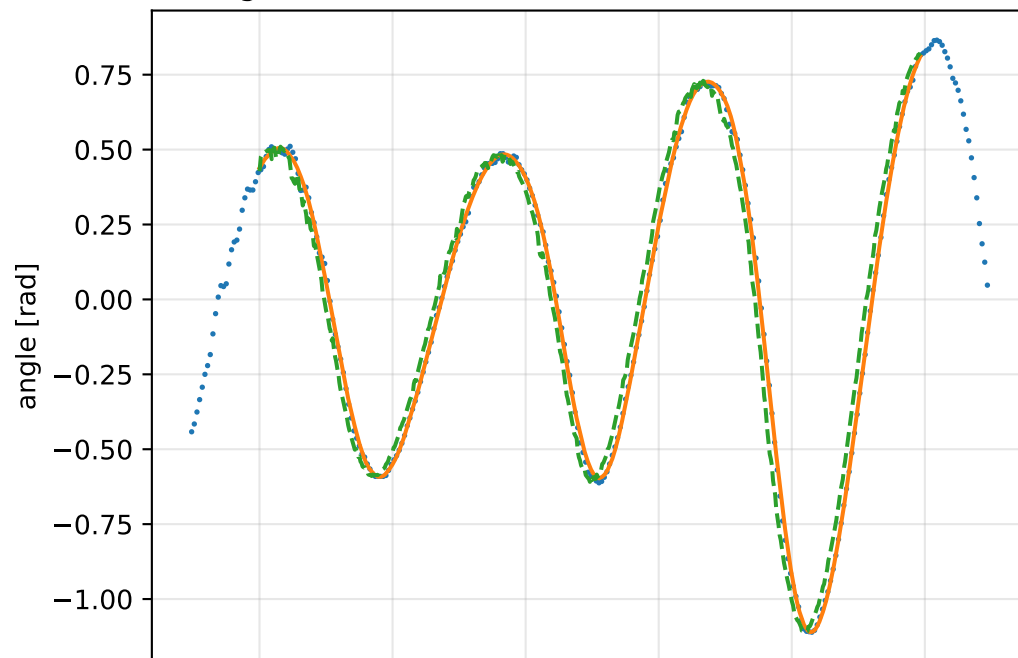


# cog\_x\_nominal: gimbal measurement audit (objective=measured\_sg)

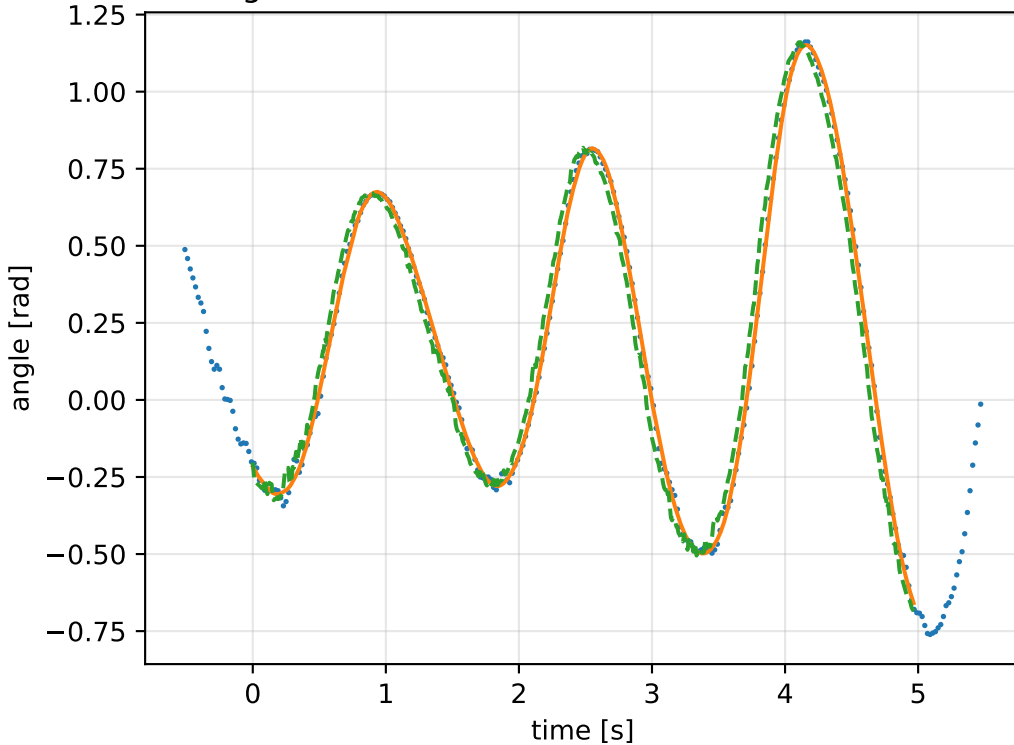
gimbal 1: RMSE=0.0759, max=0.2095 rad



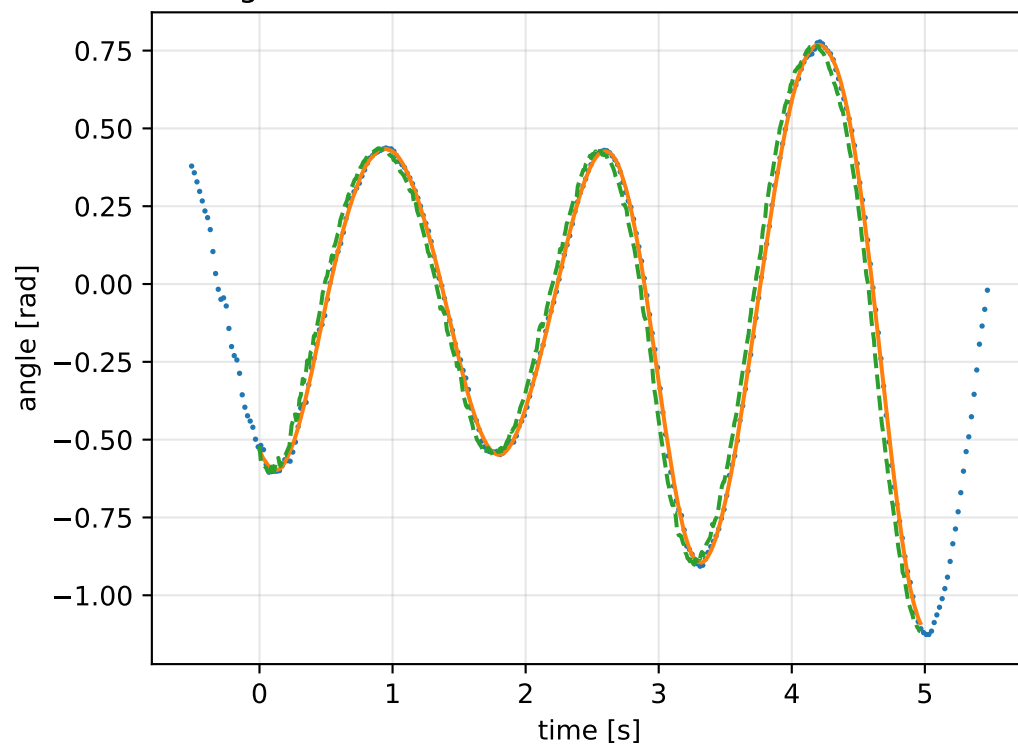
gimbal 2: RMSE=0.08099, max=0.2013 rad



gimbal 3: RMSE=0.07393, max=0.172 rad

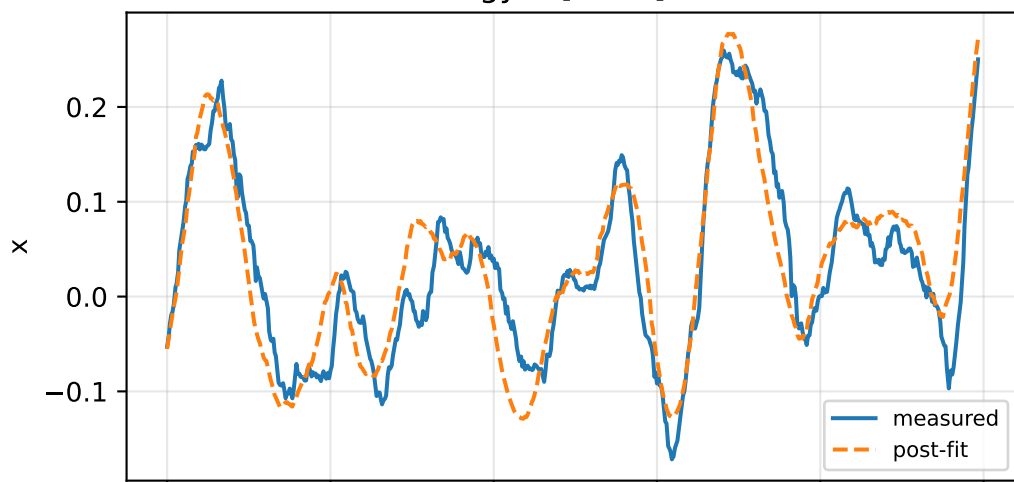


gimbal 4: RMSE=0.07131, max=0.1737 rad

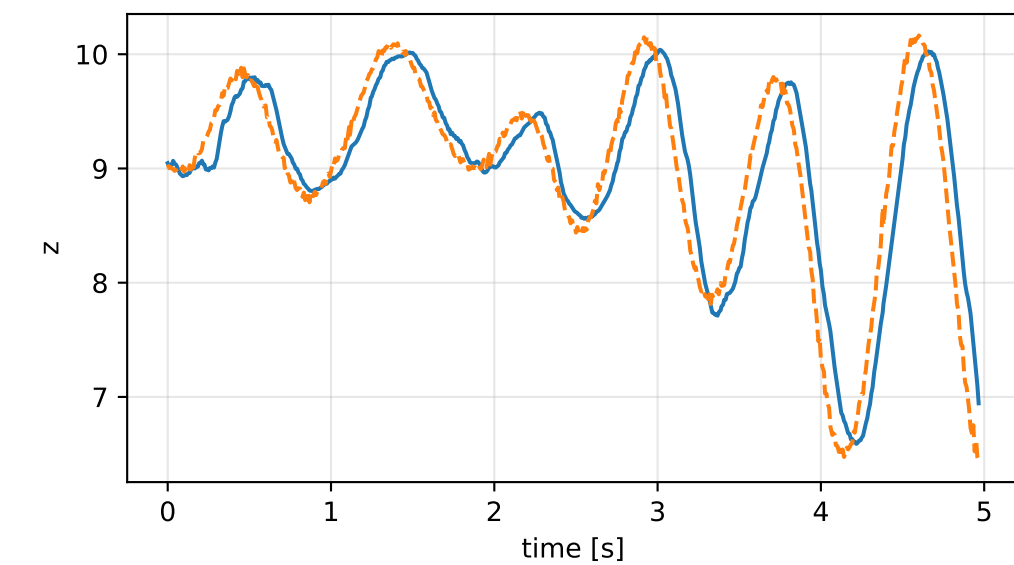
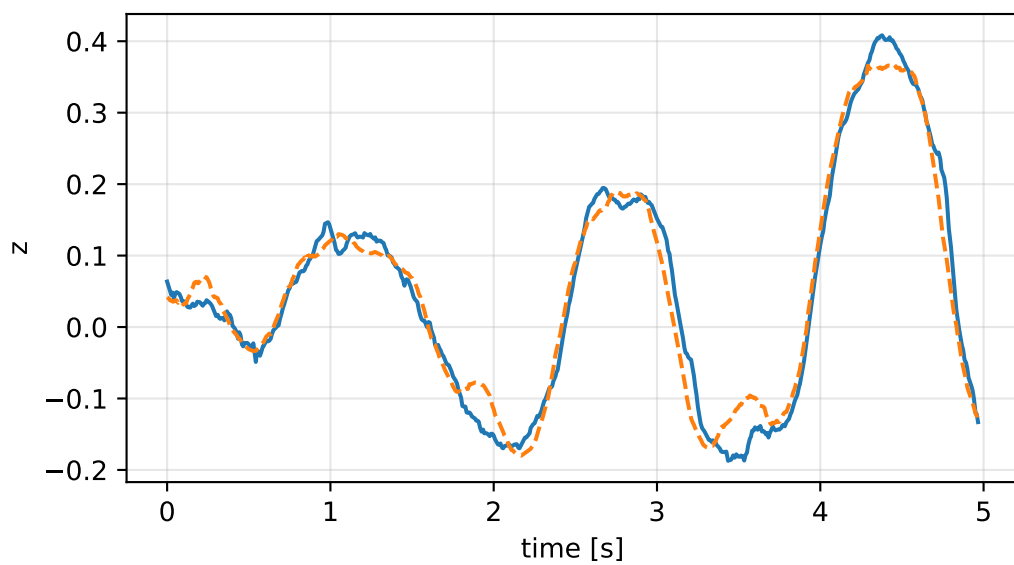
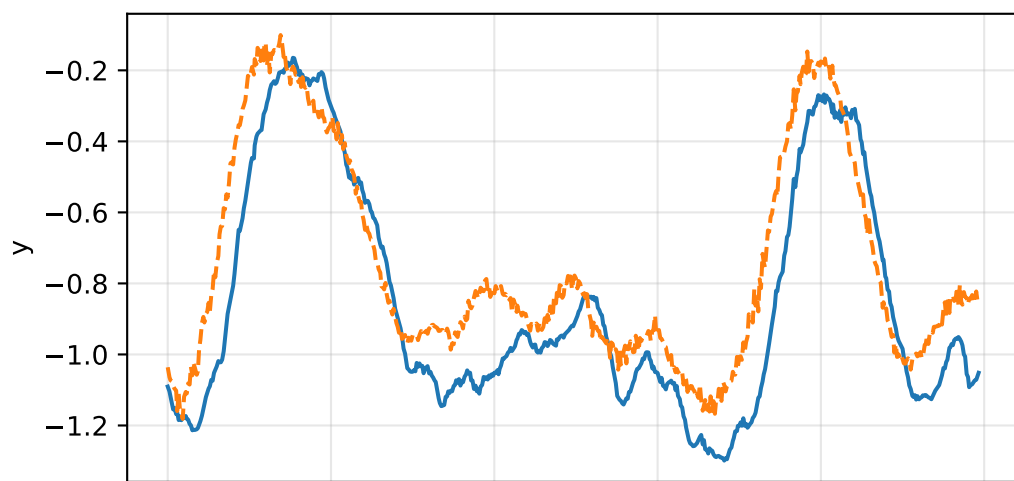
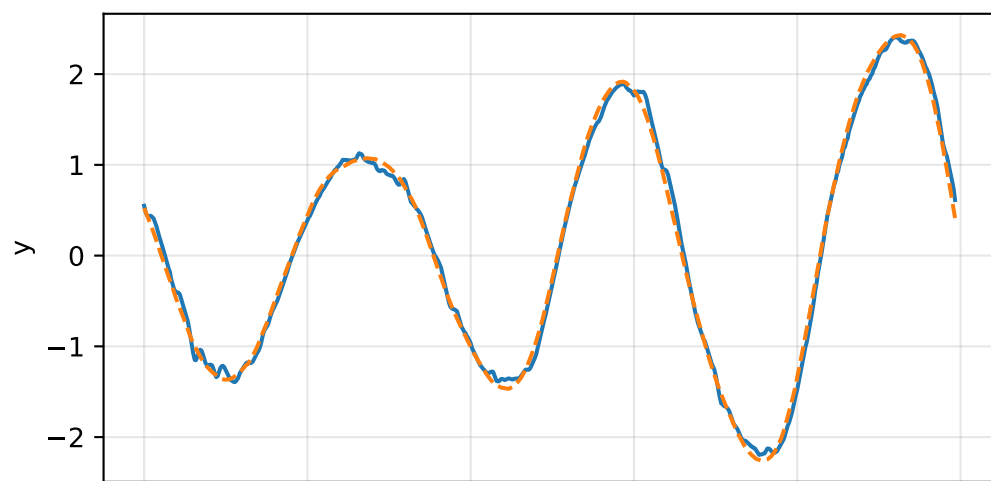
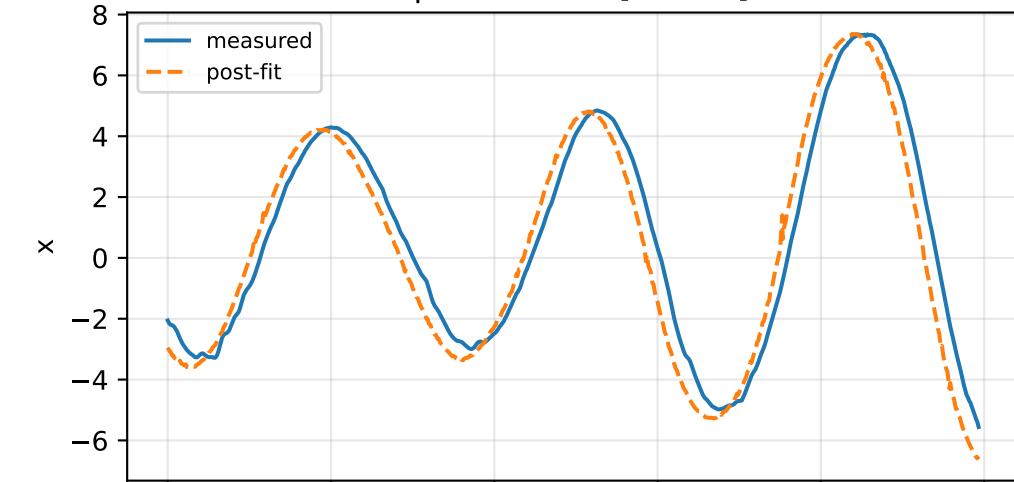


# cog\_x\_nominal: IMU comparison (diagnostic only)

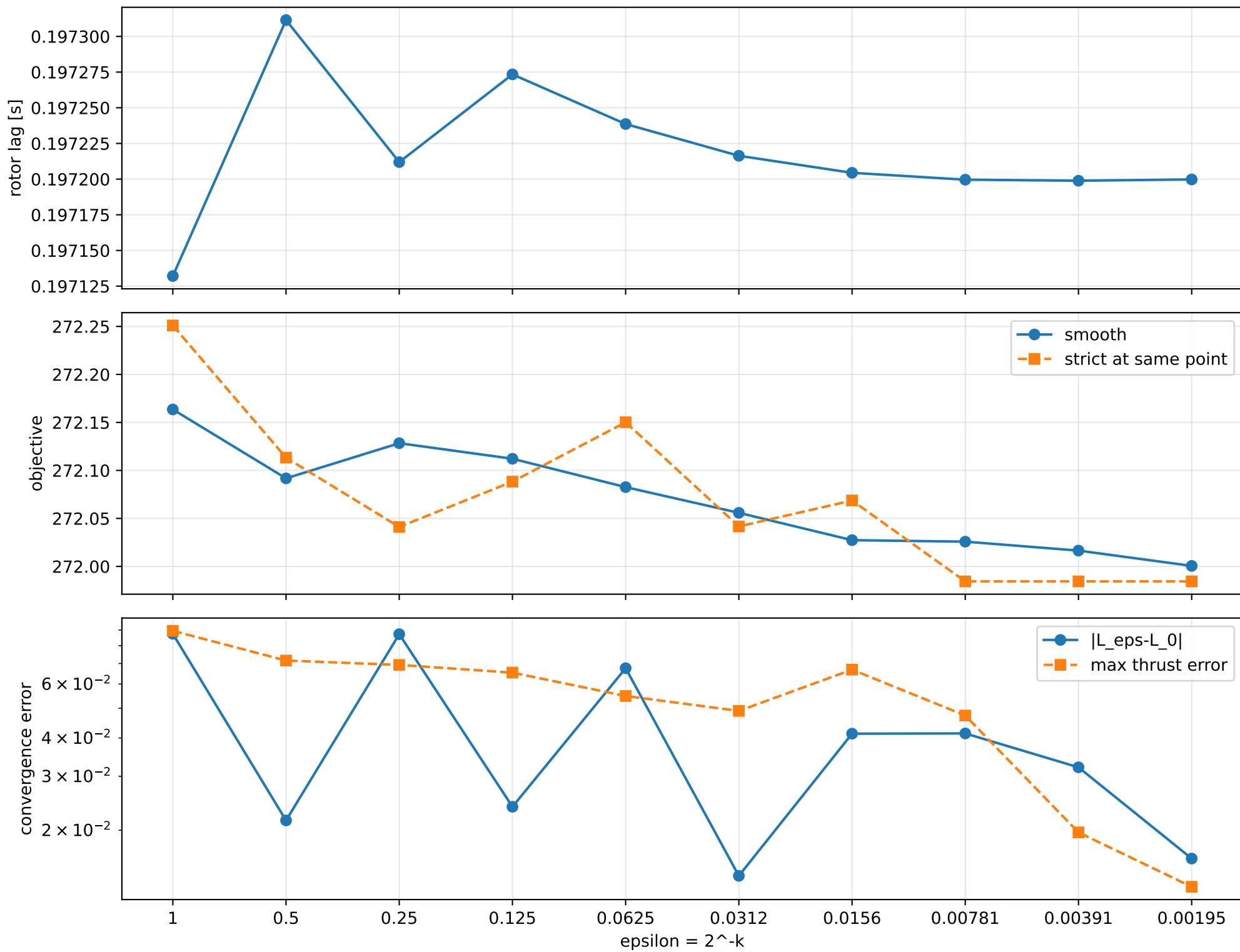
## gyro [rad/s]



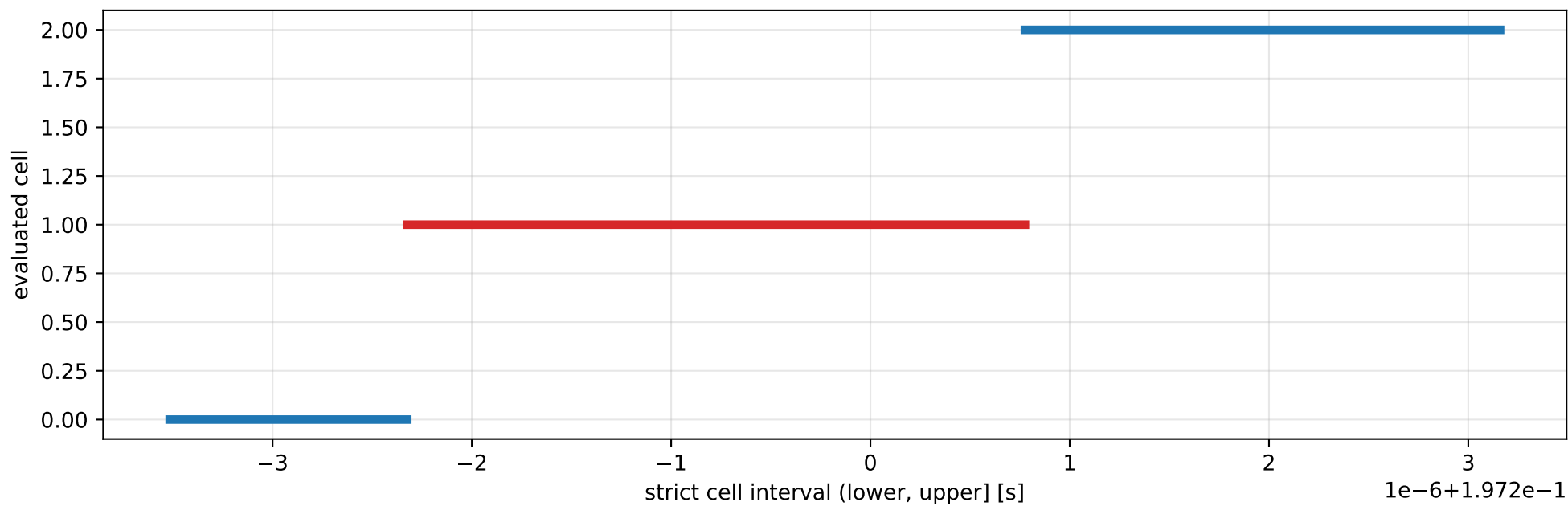
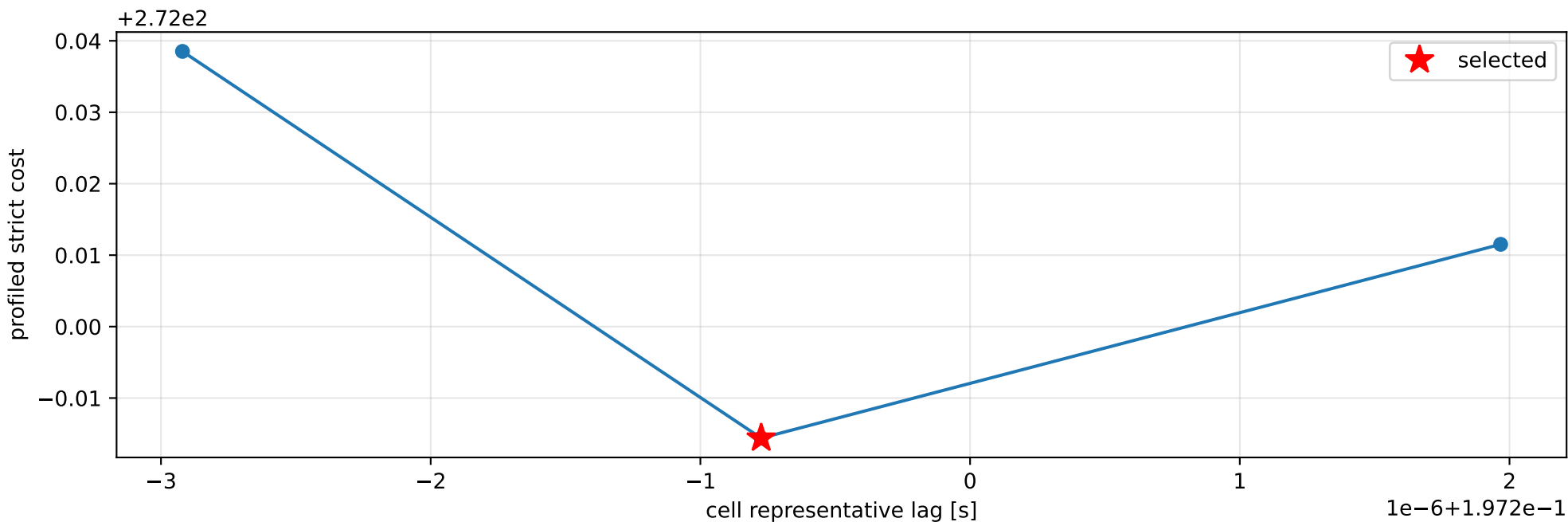
## specific force [m/s^2]



cog\_x\_nominal: smooth-to-strict continuation



cog\_x\_nominal: exact strict-ZOH lag cells



selected (0.197197676, 0.197200775] s; final smooth 0.197199757 s; data boundary=False

## cog\_x\_nominal: parameter estimate

Case: cog\_x\_nominal

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 3.34120908

inertia [kg m<sup>2</sup>]:

```
[[ 0.34354966  0.0013909 -0.06638609]
 [ 0.0013909  0.26370973 -0.00635765]
 [-0.06638609 -0.00635765  0.57107873]]
```

CoG body [m]: [-0.00202467 0.0008361 -0.04574966]

force effectiveness: [1.19717602 1.31180952 1.17055889 1.17186234]

scale-free J/m [m<sup>2</sup>]:

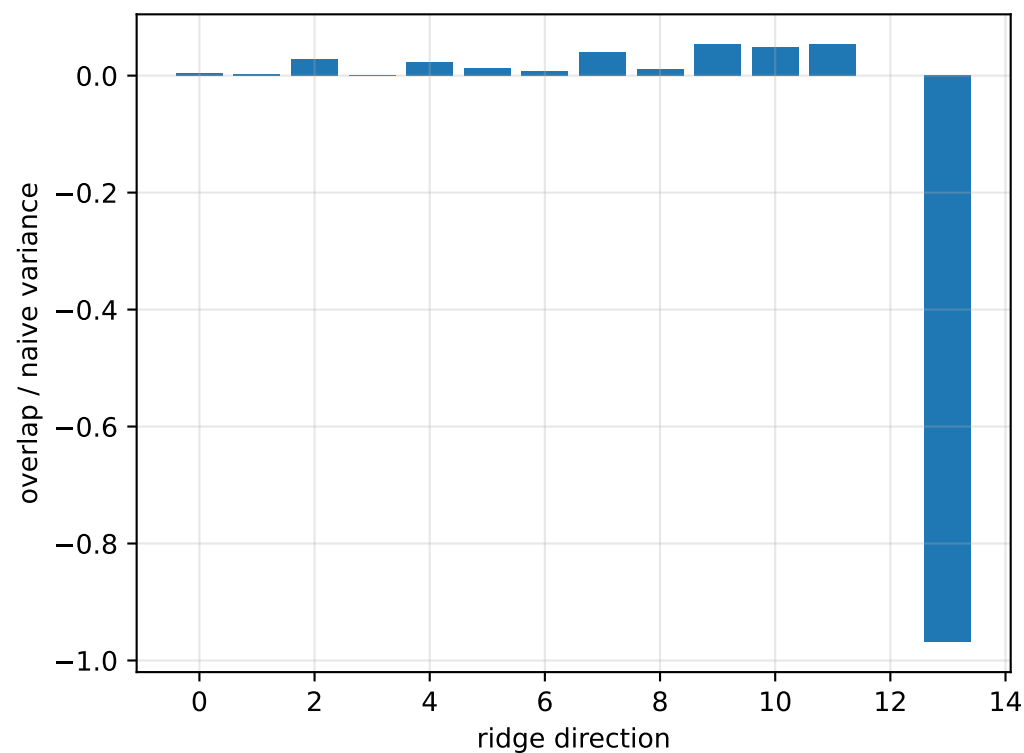
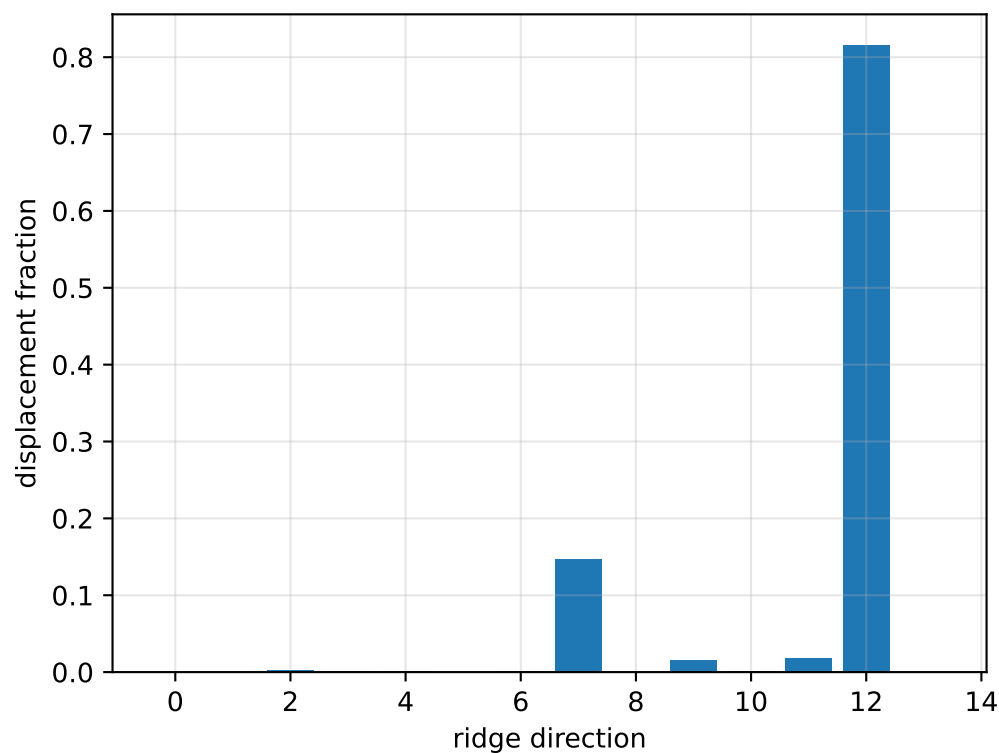
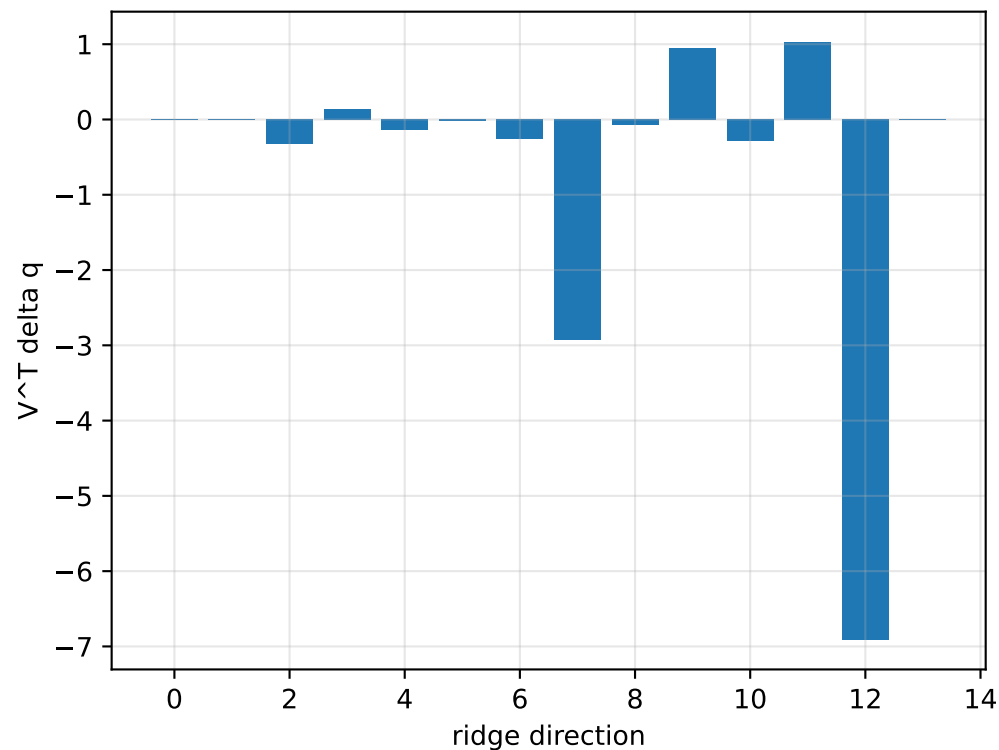
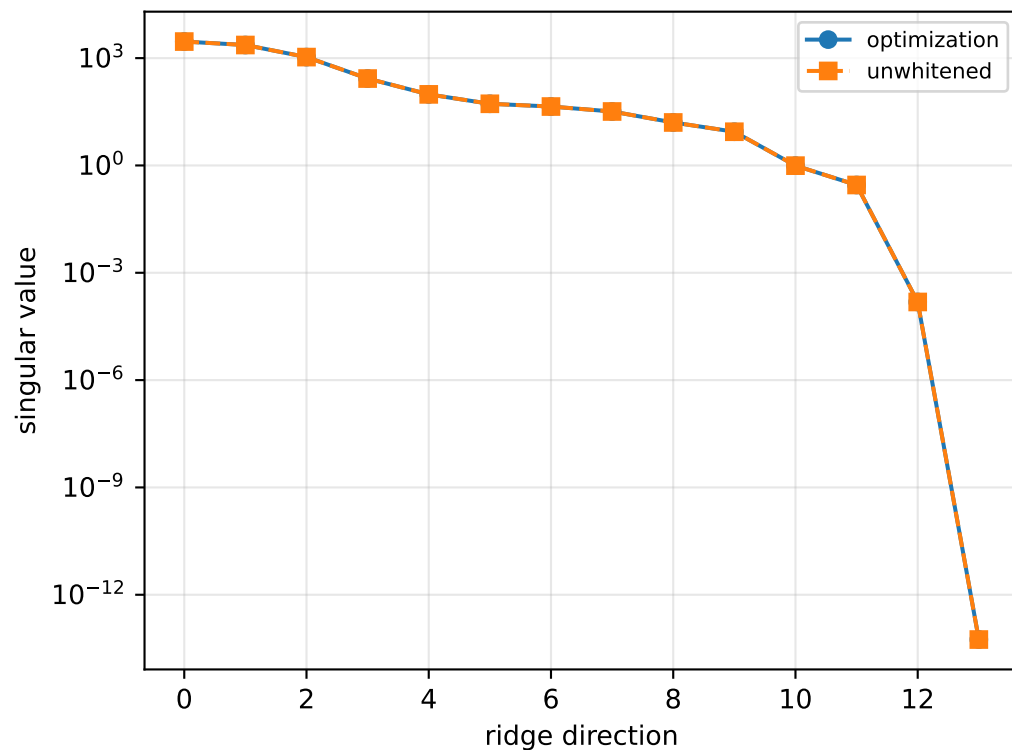
```
[[ 0.10282196  0.00041629 -0.01986888]
 [ 0.00041629  0.07892644 -0.0019028 ]
 [-0.01986888 -0.0019028  0.17091978]]
```

scale-free f/m: [0.35830623 0.39261521 0.35033991 0.35073002]

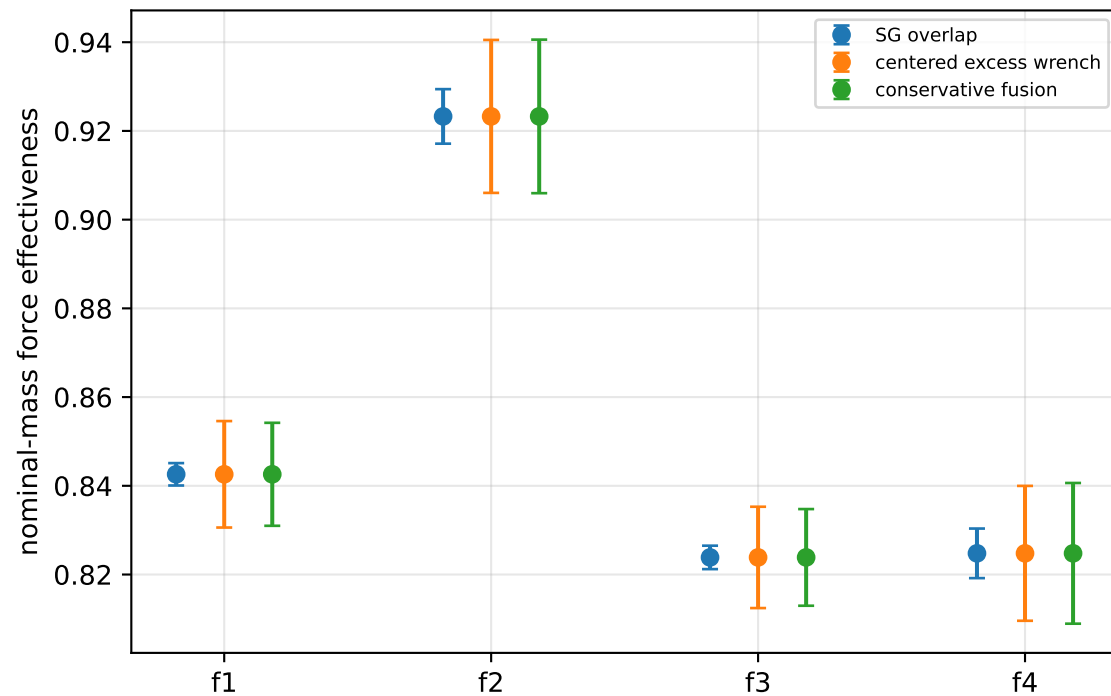
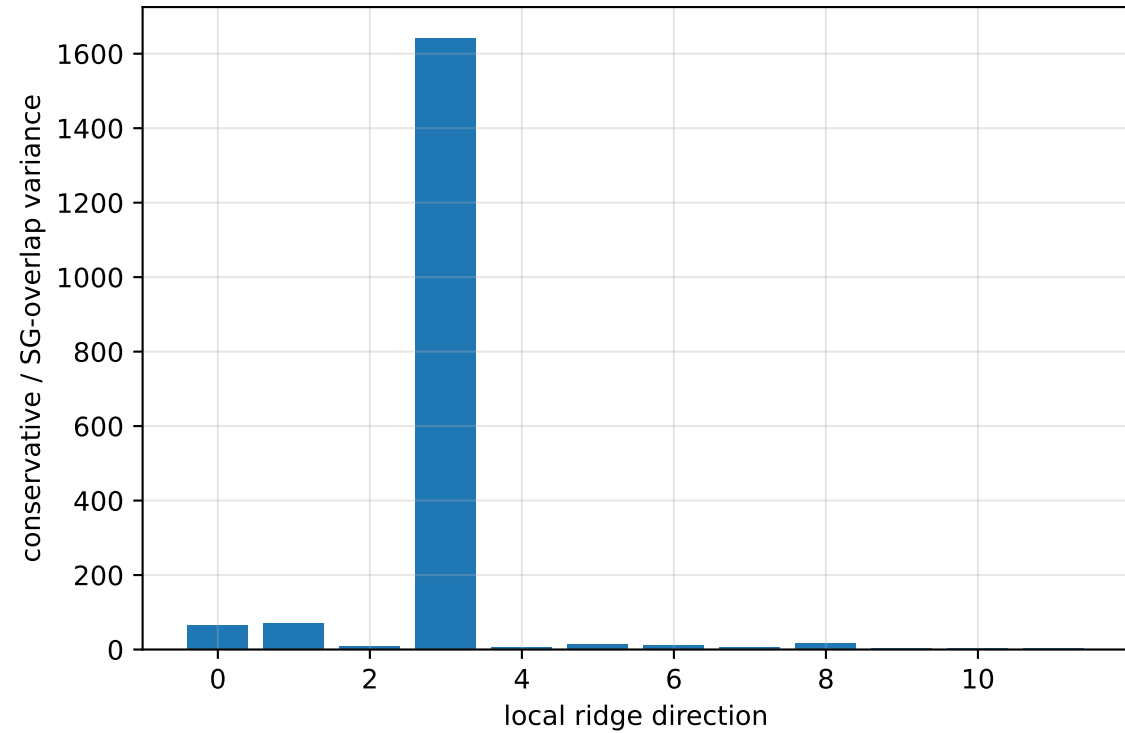
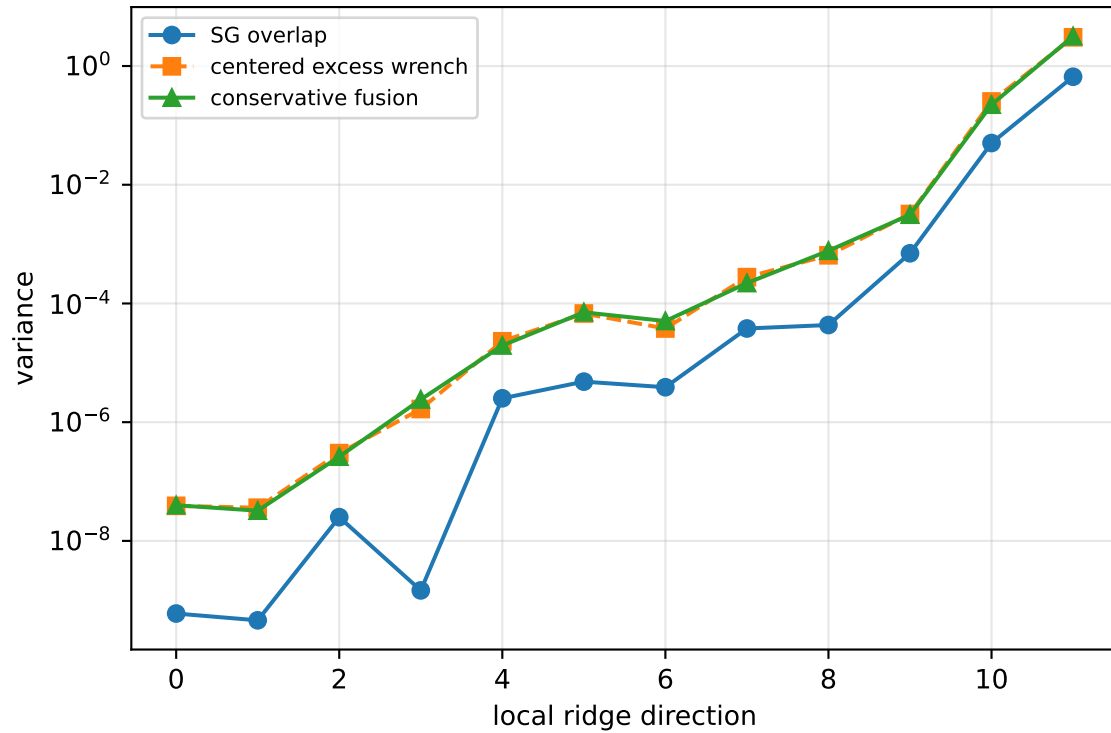
rotor lag cell: (0.197197676, 0.197200775] s

representative: 0.197199225 s

cog\_x\_nominal: ridge spectrum (rank 13, nullity 1,  $||jv||=9.27e-14$ )



## cog\_x\_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions  
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 45.43$   
wrench-to-acceleration recovery max error =  $3.83\text{e-}15$   
exact scale gauge ridge index = 13 (alignment 1)

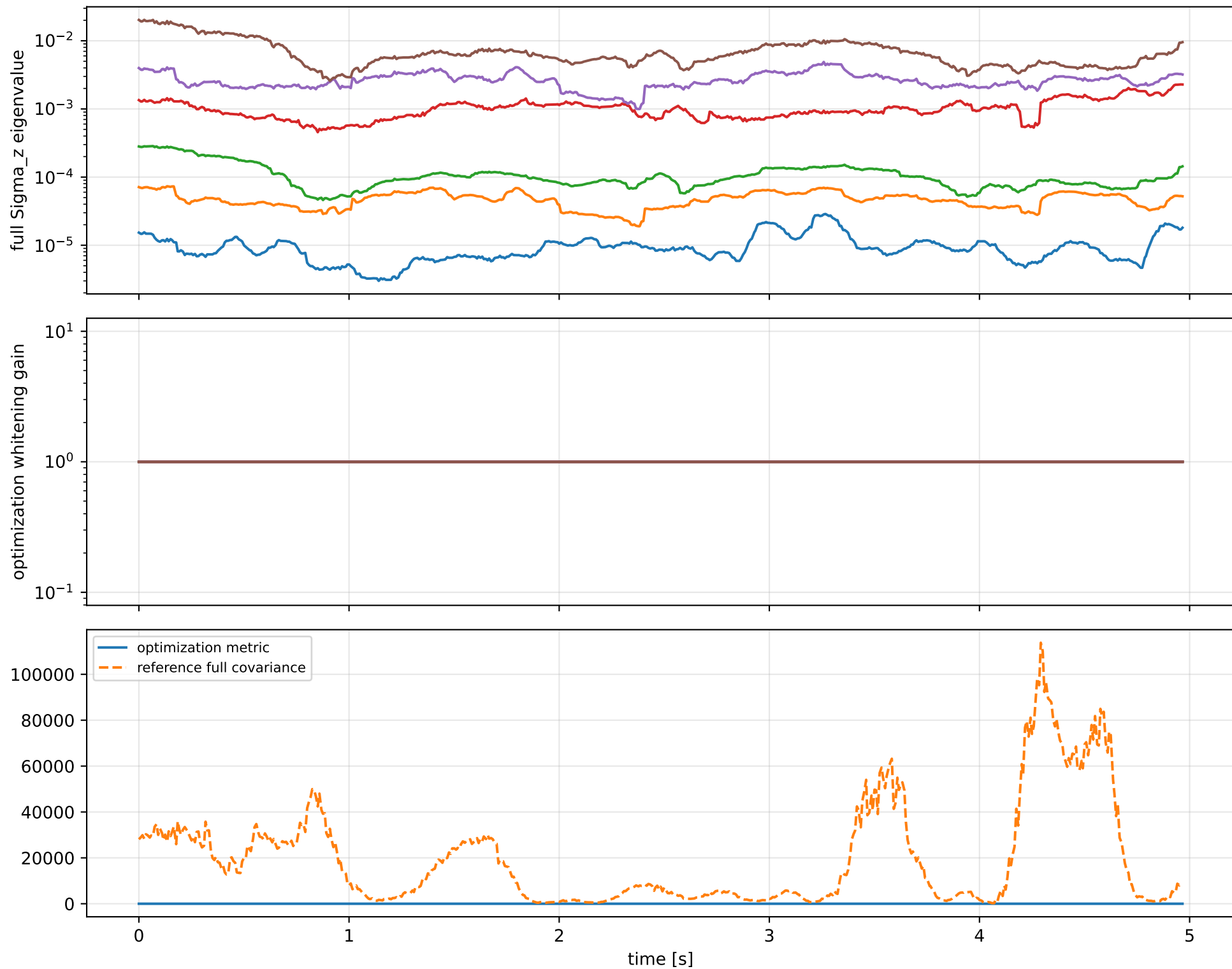
index 11: variance=3.172, ratio=4.807  
 $\log_{\text{mass}}=-0.0525$ ; second\_moment\_diag\_1=-0.366  
second\_moment\_diag\_2=-0.0242; second\_moment\_diag\_3=+0.653  
second\_moment\_offdiag\_12=+0.2; second\_moment\_offdiag\_13=-0.601  
second\_moment\_offdiag\_23=-0.155; cog\_x=+8.29e-05  
cog\_y=-0.000257; cog\_z=+3.19e-05  
log\_force\_eff\_1=-0.0522; log\_force\_eff\_2=-0.0507  
log\_force\_eff\_3=-0.0537; log\_force\_eff\_4=-0.0535

index 10: variance=0.2232, ratio=4.425  
 $\log_{\text{mass}}=+0.0125$ ; second\_moment\_diag\_1=-0.00588  
second\_moment\_diag\_2=+0.125; second\_moment\_diag\_3=-0.181  
second\_moment\_offdiag\_12=+0.879; second\_moment\_offdiag\_13=+0.187  
second\_moment\_offdiag\_23=-0.377; cog\_x=+0.00175  
cog\_y=-0.00423; cog\_z=+0.000393  
log\_force\_eff\_1=+0.0214; log\_force\_eff\_2=+0.036  
log\_force\_eff\_3=+0.00302; log\_force\_eff\_4=-0.0118

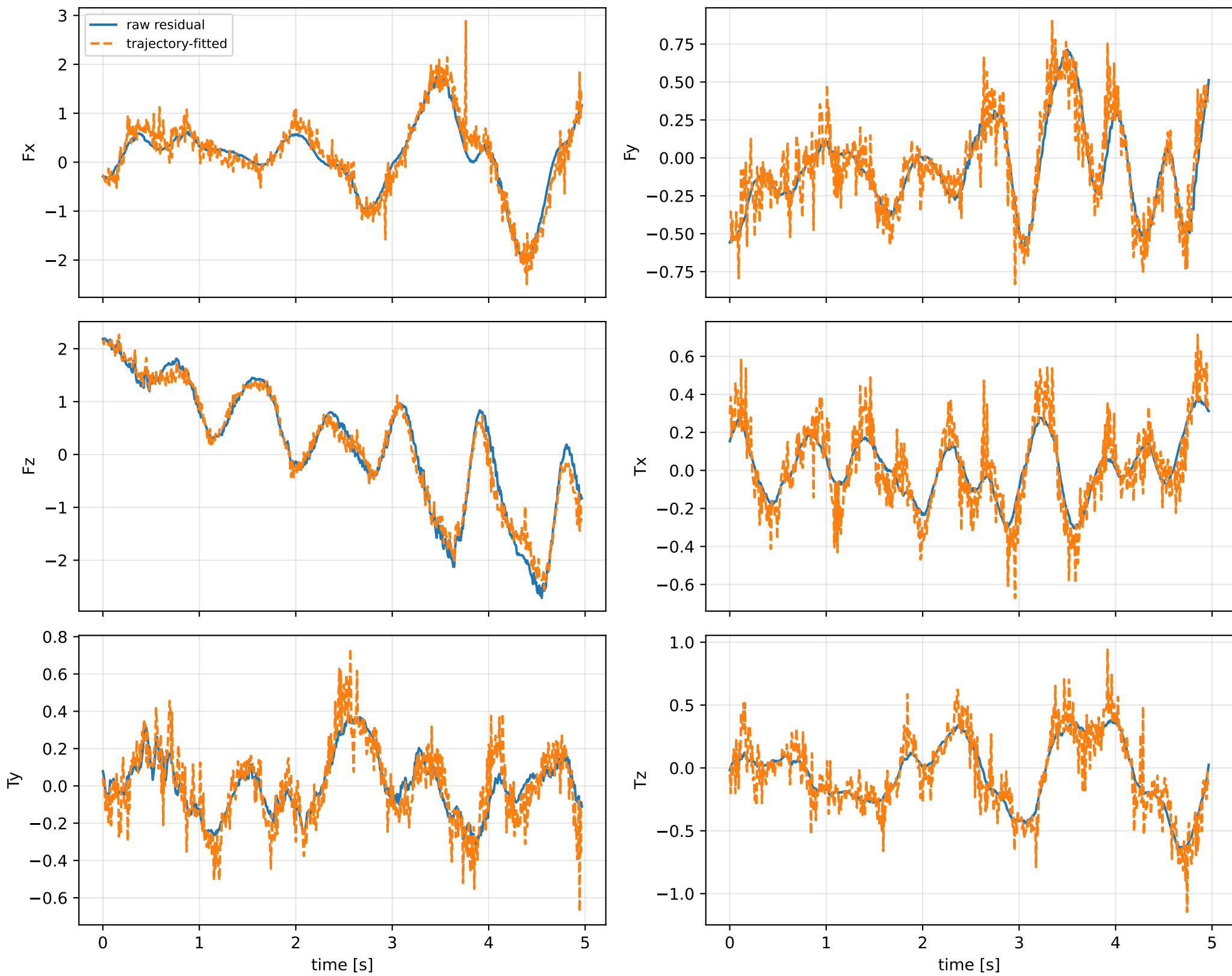
index 9: variance=0.003128, ratio=4.465  
 $\log_{\text{mass}}=-0.164$ ; second\_moment\_diag\_1=-0.0219  
second\_moment\_diag\_2=+0.911; second\_moment\_diag\_3=-0.0699  
second\_moment\_offdiag\_12=-0.0784; second\_moment\_offdiag\_13=-0.0786  
second\_moment\_offdiag\_23=+0.0975; cog\_x=+0.00477  
cog\_y=-0.00949; cog\_z=-0.00528  
log\_force\_eff\_1=-0.137; log\_force\_eff\_2=-0.115  
log\_force\_eff\_3=-0.169; log\_force\_eff\_4=-0.235

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

cog\_x\_nominal: optimization vs reference covariance



cog\_x\_nominal: wrench / closure (force max 1.31e-14, torque max 8.67e-16)



## cog\_x\_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo\_condition\_cog\_x\_nominal

role: pseudo\_conditioning\_ablation

source: /home/leus/catkin\_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo\_conditioning/scalar/cog\_x\_nominal.j

SHA256: bbfe2d2485bf27a6773ee0d77fa915418fa80f5a855431ac4e09df6ae9248b8d

data objective: 271.984386315

prior objective: 5.79382978522e-06

total objective: 271.984392109

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: cog\_x\_nominal (cog\_position\_body\_m)

components: x

target: [-0.00202471]

std: [1.e-05]

final: [-0.00202467]

error: [3.40406515e-08]

standardized residual: [0.00340407]

factor objective: 5.79382978522e-06